THE COMPLETION FUNCTOR IS AN EQUIVALENCE OVER AN AFFINE BASE

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file restricted.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: the source proves the lemma in three explicit steps (affine, separated, general), quoting the algebraization and bootstrap results for the gluing; the argument is complete

Editorial note. The label lemma-dilatations-affine names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

Allowed standard prerequisites (editorial).

- the completion functor from the chapter's construction and its full faithfulness lemma (counted earlier in this lane)
- rig-etale morphisms of formal algebraic spaces and the algebraization lemma for affine rig-etale targets
- the smash construction away from T producing formal completions of etale covers
- etale equivalence relations and the bootstrap theorem producing quotient algebraic spaces
- compatibility of completion with fibre products and quotient sheaves

Source reference. Original theorem 04S6: sources/stacks-chow/tags.txt, source lines 6209–6209. Prerequisite pointer invoked at proof line(s) 6999; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original equation 0880: sources/stacks-restricted/pinned-original-source.tex, source lines 6766–6766. Prerequisite pointer invoked at proof line(s) 6948; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original definition 0AIM: sources/stacks-chow/tags.txt, source lines 13270–13270. Prerequisite pointer invoked at proof line(s) 6980; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 0AR4: sources/stacks-restricted/pinned-original-source.tex, source lines 6406–6406. Prerequisite pointer invoked at proof line(s) 6974; it names a standard fact of the same chapter that the counted argument

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uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 0GDR: sources/stacks-restricted/pinned-original-source.tex, source lines 6832–6832. Prerequisite pointer invoked at proof line(s) 6964; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 024E: sources/stacks-restricted/pinned-original-source.tex, source lines 6517–6517. Prerequisite pointer invoked at proof line(s) 6995; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 0AR9: sources/stacks-restricted/pinned-original-source.tex, source lines 6554–6554. Prerequisite pointer invoked at proof line(s) 6959; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original section 01JD: sources/stacks-chow/tags.txt, source lines 2010–2010. Prerequisite pointer invoked at proof line(s) 6957; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Lemma 0.1. *In the situation above. Assume X is affine. Then the functor (0880) is an equivalence.*

Proof. We already know the functor is fully faithful, see Lemma 0GDR. Essential surjectivity. Let $g : W \to X_{/T}$ be a morphism of formal algebraic spaces with W locally Noetherian and g rig-étale. We will prove W is in the essential image in a number of steps.

Step 1: W is an affine formal algebraic space. Then we can find $U \to X$ of finite type and étale over $X \setminus T$ such that $U_{/T}$ is isomorphic to W, see Lemma 0AR4. Thus we see that W is in the essential image by Lemma 0AR9.

Step 2: W is separated. Choose $\{W_i \to W\}$ as in Formal Spaces, Definition 0AIM. By Step 1 the formal algebraic spaces W_i and $W_i \times_W W_j$ are in the essential image. Say $W_i = (X'_i)_{/T}$ and $W_i \times_W W_j = (X'_{ij})_{/T}$. By fully faithfulness we obtain morphisms $t_{ij} : X'_{ij} \to X'_i$ and $s_{ij} : X'_{ij} \to X'_j$ matching the projections $W_i \times_W W_j \to W_i$ and $W_i \times_W W_j \to W_j$. Consider the structure

$$R = \coprod X'_{ij}, \quad V = \coprod X'_i, \quad s = \coprod s_{ij}, \quad t = \coprod t_{ij}$$

(We can't use the letter U as it has already been used.) Applying Lemma 024E we find that $(t, s) : R \to V \times_X V$ defines an étale equivalence relation on V over X. Thus we can take the quotient $X' = V/R$ and it is an algebraic space, see Bootstrap, Theorem 04S6. Since completion commutes with fibre products and taking quotient sheaves, we find that $X'_{/T} \cong W$ as formal algebraic spaces over $X_{/T}$.

Step 3: W is general. Choose $\{W_i \to W\}$ as in Formal Spaces, Definition 0AIM. The formal algebraic spaces W_i and $W_i \times_W W_j$ are separated. Hence by Step 2 the formal algebraic spaces W_i and $W_i \times_W W_j$ are in the essential image. Then we argue exactly as in the previous paragraph to see that W is in the essential image as well. This concludes the proof. □